

Lesson 1

Code · Git · GitHub · Vercel

How files become websites

What you'll understand after this lesson

- What a website actually *is* at the file level
- Why filenames like **index.html** matter
- What Git and GitHub are for
- What Vercel does and why you need it
- How all three connect into one pipeline

1 What is a website, really?

A website is just **text files sitting on a computer** that is always on and connected to the internet. When you type a URL into a browser, you're asking that computer to send you its files. Your browser reads them and draws the page.

The three file types every web page uses:

File type	What it does	Analogy
HTML (.html)	Structure — the content itself	The walls of a building
CSS (.css)	Style — colours, fonts, layout	Paint and furniture
JavaScript (.js)	Behaviour — interactivity, logic	Doors that open, lights that switch

■ Your German lesson app in one sentence

index.html contains HTML, CSS, and JavaScript all in one file. When Vercel serves it to your browser, the browser reads the HTML for structure, applies the CSS for the dark theme, and runs the JavaScript to call the AI and render lessons.

Why the file must be called index.html

Web servers have a 40-year-old convention: when someone visits a URL with no filename at the end (like **project-3gbvw.vercel.app/**), the server looks for a file called **index.html** in that folder. It's the default front door. Every hosting platform — Vercel, Netlify, GitHub Pages, Apache, Nginx — follows this rule.

```
project-3gbvw.vercel.app/ ← no filename → server looks for index.html
project-3gbvw.vercel.app/index.html ← explicit, same result
project-3gbvw.vercel.app/about.html ← different page, explicit filename required
```

That's why when you uploaded `german_daily_lesson.html`, the site was blank — Vercel served the empty `index.html` instead. Renaming to `index.html` fixed it immediately.

2 What is Git?

Git is a **version control system**. It tracks every change you make to your files over time, so you can always go back to any previous state. Think of it as a time machine for your code.

The problem Git solves

Without Git, people save files like this:

```
essay_v1.docx
essay_v2.docx
essay_final.docx
essay_FINAL_real.docx
essay_FINAL_real_USE_THIS_ONE.docx
```

Git does this automatically and intelligently for every file in your project.

Key Git concepts

Repository (repo)

The folder Git is tracking. Contains all your files + their full history.

Commit

A snapshot of your files at a moment in time. Like pressing "Save" but forever.

Branch

A copy of your repo to experiment on, without affecting the main version.

Main branch

The official, live version of your code.

Push

Upload your local commits to the cloud (GitHub).

Pull

Download latest changes from the cloud to your local machine.

3 What is GitHub?

GitHub is **cloud storage for Git repositories**, with a web interface. It's not the same as Git — Git is the tool, GitHub is where you store your Git repos online. Alternatives exist (GitLab, Bitbucket) but GitHub is by far the most widely used.

Your repo: github.com/lld-gif/deutsch-daily

This repo contains two files: `index.html` (your lesson app) and `api/chat.js` (the proxy). Every time you commit a change to `main`, Vercel sees it and redeploys automatically.

What a commit history looks like

Every change you made to the German lesson app is permanently recorded:

Commit	Message	Who	When
8d97c6b	Add files via upload	lld-gif	Day 1
57e4bfd	Update index.html	lld-gif	Day 1
a3917ad	Route API calls through /api/chat proxy	lld-gif	Day 1
77959e4	Create chat.js — added api	lld-gif	Day 1
69a22d9	Update chat.js	lld-gif	Day 1
e468ba0	Update index.html — updated	lld-gif	Day 2
68450e1	Update index.html	lld-gif	Day 2
d12ac6a	Track used lessons to prevent repeats	lld-gif	Day 2

4 What is Vercel?

Vercel is a **hosting platform**. It takes your files from GitHub and puts them on servers around the world so anyone with a browser can visit your URL. It handles everything you'd otherwise need to set up yourself: servers, SSL certificates, global CDN, custom domains, and automatic redeployment when you push new code.

What Vercel gives you automatically

- **A permanent URL** — `project-3gbvw.vercel.app` never changes
- **HTTPS/SSL** — the padlock in your browser, free and automatic
- **Global CDN** — your files are cached in data centres worldwide so it loads fast everywhere
- **Auto-deploy** — every push to the main branch redeploys in ~20 seconds
- **Serverless functions** — your `api/chat.js` runs server-side without a dedicated server
- **Preview deployments** — every branch gets its own temporary URL for testing

Static vs Dynamic — what you have

	Static site (what you have)	Dynamic app
Files served	HTML/CSS/JS only	HTML + server-side code
Backend	None — browser does work	Database, user accounts, etc.
Hosting cost	Free on Vercel	Usually paid
Deploy time	~20 seconds	Minutes
Example	Your German lesson app	Twitter, Gmail, Airbnb

5 The full pipeline

This is exactly what happens every time you update your German lesson app:

Step 1 — Edit index.html

You open the file in the GitHub web editor, make a change (or paste a new version), and click Commit.

Step 2 — Git records a commit

GitHub stores a permanent snapshot: what changed, who changed it, when. Commit hash: d12ac6a.

Step 3 — Vercel detects the push

Vercel watches your repo via a webhook. The moment main branch is updated, Vercel is notified.

Step 4 — Vercel builds and deploys

Vercel copies your files, builds the serverless function (api/chat.js), and distributes everything globally.

Step 5 — Your URL is live

project-3gbvw.vercel.app now serves the new version. Takes ~20 seconds from commit to live.

■ The key insight

You never interact with a server directly. You edit a file → push to GitHub → Vercel handles everything else. This is modern web development for the vast majority of projects.

Practical Exercise

This exercise takes about 10 minutes and requires no coding. You'll make a real change to your live German lesson app.

What you're doing

You'll change the header title of your lesson app from **■■ Tageslektion** to **■■ Tageslektion — Guten Morgen!**, deploy it, and verify it's live.

Step-by-step

Step 1: Open your repo

```
Go to: github.com/lld-gif/deutsch-daily
Click on index.html in the file list.
Click the pencil icon (Edit) in the top right of the file view.
```

Step 2: Find the title

```
Press Cmd+F (or use the browser's built-in search if available) to find:
■■ Tageslektion
It's around line 97 in the HTML.
```

Step 3: Make the change

```
Change it to:
■■■■■■■■ Tageslektion — Guten Morgen!
```

Step 4: Commit

```
Scroll to the top and click "Commit changes..."
Write a commit message: Update page title
Click "Commit changes" (green button).
```

Step 5: Wait 20 seconds, then verify

Open `project-3gbvw.vercel.app` in your browser.
Hard refresh: `Cmd+Shift+R` (clears cache).
You should see your new title.

Step 6: Check your commit history

Go back to `github.com/lld-gif/deutsch-daily`
Click '9 Commits' (or however many there are now).
Your new commit should be at the top with timestamp.

✓ What you just proved

You edited a file on GitHub, Git recorded the change, Vercel detected it and redeployed, and your live URL updated — all automatically. That's the entire pipeline in action.

Checkpoint questions

- → What would happen if you renamed `index.html` to `myapp.html` and visited your Vercel URL?
- → If you made a mistake in your commit, how would you fix it? (Hint: make another commit.)
- → What's the difference between Git and GitHub?
- → Why does Vercel redeploy automatically when you push to GitHub?